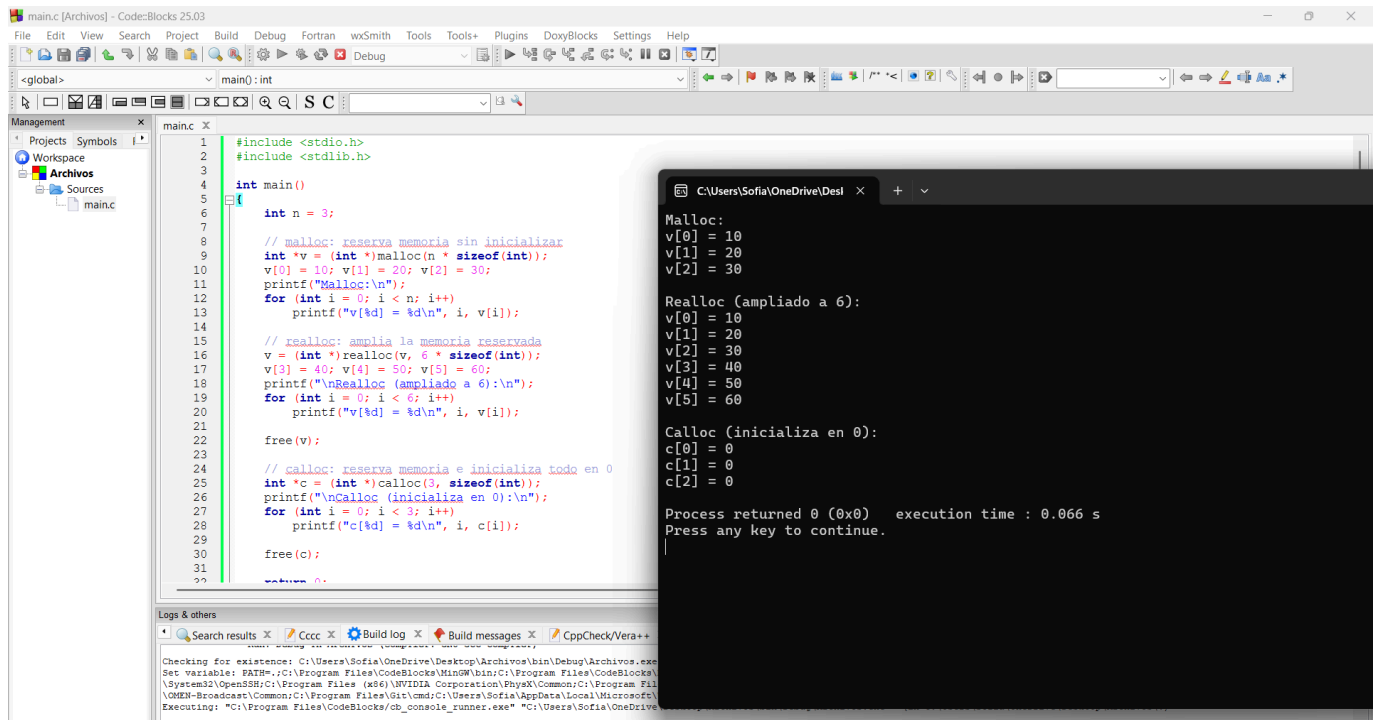


Memoria - Documentación

/////Captura de funcionamiento del código

- Código memoria_dinamica.c



The screenshot displays the Code::Blocks IDE with the source code of `memoria_dinamica.c` in the main editor. The code demonstrates dynamic memory management using `malloc`, `realloc`, and `calloc`. It includes comments in Spanish explaining each step. A terminal window in the foreground shows the program's output, which includes the memory addresses and values for arrays `v` and `c` at different stages of execution.

```
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int n = 3;

    // malloc: reserva memoria sin inicializar
    int *v = (int *)malloc(n * sizeof(int));
    v[0] = 10; v[1] = 20; v[2] = 30;
    printf("Malloc:\n");
    for (int i = 0; i < n; i++)
        printf("v[%d] = %d\n", i, v[i]);

    // realloc: amplia la memoria reservada
    v = (int *)realloc(v, 6 * sizeof(int));
    v[3] = 40; v[4] = 50; v[5] = 60;
    printf("\nRealloc (ampliado a 6):\n");
    for (int i = 0; i < 6; i++)
        printf("v[%d] = %d\n", i, v[i]);

    free(v);

    // calloc: reserva memoria e inicializa toda en 0
    int *c = (int *)calloc(3, sizeof(int));
    printf("\nMalloc (inicializa en 0):\n");
    for (int i = 0; i < 3; i++)
        printf("c[%d] = %d\n", i, c[i]);

    free(c);
}
```

Execution output from the terminal:

```
Malloc:
v[0] = 10
v[1] = 20
v[2] = 30

Realloc (ampliado a 6):
v[0] = 10
v[1] = 20
v[2] = 30
v[3] = 40
v[4] = 50
v[5] = 60

Malloc (inicializa en 0):
c[0] = 0
c[1] = 0
c[2] = 0

Process returned 0 (0x0)   execution time : 0.066 s
Press any key to continue.
```